Mathematics practice booklet: Grade 4 expectations

Grade 4 · Practice booklet

Name: _____

What this booklet is

This is general practice written for Mathematics at Grade 4 level, following the Common Core State Standards for Mathematics standards for that grade. It is not made from your child's MAP questions, and it does not replace the classroom. Use it as a guide to what a child is expected to know and do at this level.

How to use it

- Do one section at a time, about 10 minutes, 3 days a week. Stop while it is still enjoyable.
- Read each question all the way through, twice, before answering. Underline the key words and say what is being asked.
- Take your time. A careful, correct answer is worth more than a fast one.
- Use scrap paper and show the working. Then check the answer before moving on.
- Ask your child to explain how they got an answer, even when it is right.
- If a section feels too hard, do the section before it again, or ask the teacher. That is part of learning, not a problem.

Section 1: Patterns, equations and algebra thinking

Standards practised: 4.OA.A.1, 4.OA.A.2, 4.OA.A.3, 4.OA.B.4, 4.OA.C.5

A worked example

A ribbon is 3 times as long as a string. The string is 14 cm. How long is the ribbon? Write the equation.

1. 3 times as long means multiply: 3 x 14.
2. 3 x 10 = 30 and 3 x 4 = 12, so 30 + 12 = 42.
3. Equation: r = 3 x 14 = 42.

42 cm

Questions

1. Write as a multiplication: 24 is 4 times as many as 6.

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2. Maryam has 9 stickers. Noor has 5 times as many. How many stickers does Noor have?

   A. 14

   B. 4

   C. 45

   D. 54

3. A shirt costs AED 36 and a bag costs 3 times as much. How much do they cost together?

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4. 47 students go on a trip to see falcons and each bus holds 12 students. How many buses are needed?

   A. 4

   B. 3

   C. 11

   D. 35

5. List all the factor pairs of 24.

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6. Which of these numbers is prime?

   A. 27

   B. 29

   C. 21

   D. 33

7. Hind packs date cookies into boxes of 9. She has 36 cookies. Can she fill the boxes exactly with none left over? Explain.

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8. Continue the pattern: 5, 11, 17, 23, __, __. What is the rule?

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9. A pattern starts at 3 and the rule is multiply by 2. What are the first four numbers?

   A. 3, 5, 7, 9

   B. 3, 6, 9, 12

   C. 3, 6, 12, 18

   D. 3, 6, 12, 24

10. Ali scores 8 more points than Sami, who scores 3 times as many as Hind. Hind scores 6. How many does Ali score?

Section 2: Shapes, measuring and space

Standards practised: 4.MD.A.1, 4.MD.A.2, 4.MD.A.3, 4.MD.C.5, 4.MD.C.6, 4.MD.C.7, 4.G.A.1, 4.G.A.2, 4.G.A.3

A worked example

A rectangular football pitch for children is 40 m long and 25 m wide. Find the perimeter and the area.

1. Perimeter = 2 x (40 + 25) = 2 x 65 = 130 m.
2. Area = 40 x 25 = 1,000 square metres.

Perimeter 130 m; area 1,000 square metres

Questions

1. A rectangular mat in a model of a souq stall has an area of 56 square cm and a length of 8 cm. What is its width?

   A. 48 cm

   B. 448 cm

   C. 7 cm

   D. 64 cm

2. Convert: 3 km = ___ m

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3. The film lasts 2 hours 30 minutes. How many minutes is that?

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4. A rope is 2 m 40 cm and another is 1 m 85 cm. How much longer is the first, in cm?

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5. Khalid and his cousins walk once all the way round a rectangular park that is 120 m long and 80 m wide. How far do they walk?

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6. The minute hand of a clock moves from 12 to 3. Through what angle does it turn?

   A. 180 degrees

   B. 90 degrees

   C. 45 degrees

   D. 30 degrees

7. Two angles sit side by side on a straight line. One is 120 degrees. What is the other?

   A. 240 degrees

   B. 90 degrees

   C. 180 degrees

   D. 60 degrees

8. How many lines of symmetry does a rectangle that is not a square have?

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9. A triangle has angles of 90, 45 and 45 degrees. What kind of triangle is it?

   A. Right triangle

   B. Acute triangle

   C. Obtuse triangle

   D. Equilateral triangle

10. The two tracks of the metro line run side by side along the road and never meet. Are they parallel or perpendicular?

Section 3: Data, graphs and chance

Standards practised: 4.MD.B.4

A worked example

A line plot shows rope lengths in quarters of a metre: 1/4 m has 3 dots, 2/4 m has 4 dots, 3/4 m has 1 dot. What is the total length of the 2/4 m ropes?

1. There are 4 dots at 2/4.
2. 4 x 2/4 = 8/4.
3. 8/4 is the same as 2 metres.

2 metres

Questions

1. A line plot shows rope lengths: 3 dots at 1/4 m, 4 dots at 2/4 m and 1 dot at 3/4 m. How many ropes are there in all?

   A. 4

   B. 8

   C. 3

   D. 6

2. In the same line plot (3 dots at 1/4 m, 4 dots at 2/4 m, 1 dot at 3/4 m), which length is the most common?

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3. In that line plot, how much longer is the longest rope than the shortest?

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4. Hamdan makes a line plot of ropes in quarters of a metre: 2 dots at 1/4 m and 3 dots at 2/4 m. What is the total length of all these ropes?

   A. 5/4 m

   B. 5/8 m

   C. 2 m

   D. 8/8 m

5. A line plot of rainfall in centimetres, measured to the nearest half, has 3 dots at 1/2 and 2 dots at 1. What is the total?

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6. A line plot shows ribbons: 2 dots at 1/4 m, 3 dots at 1/2 m and 1 dot at 3/4 m. How many ribbons are longer than 1/4 m?

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7. A line plot shows cups of karak: 3 dots at 1/8 litre and 2 dots at 3/8 litre. How much karak is there in all?

   A. 5/8 litre

   B. 9/16 litre

   C. 1 litre

   D. 1 and 1/8 litres

8. At a family gathering 12 people chose karak, 9 chose juice and 6 chose water. How many more people chose karak than water?

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9. In a survey of 27 people, 12 chose karak, 9 chose juice and 6 chose water. What fraction chose juice, in simplest form?

   A. 1/3

   B. 3/9

   C. 9/18

   D. 1/9

10. Seven plants measure 1/2, 1/2, 3/4, 3/4, 3/4, 1, and 1 and 1/4 of a metre. How many measure 3/4 m?

Section 4: Numbers and calculating

Standards practised: 4.NBT.A.1, 4.NBT.A.2, 4.NBT.A.3, 4.NBT.B.4, 4.NBT.B.5, 4.NBT.B.6, 4.NF.A.1, 4.NF.A.2, 4.NF.B.3, 4.NF.B.4, 4.NF.C.5, 4.NF.C.6, 4.NF.C.7

A worked example

What is 3/8 + 2/8? What is 3/4 as a fraction with 8 as the denominator?

1. The denominators match, so add the numerators: 3 + 2 = 5, giving 5/8.
2. For 3/4: multiply the top and bottom by 2: 6/8.

5/8 and 6/8

Questions

1. Write in words: 45,208

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2. A solar park made 36,482 units of electricity in a month. Rounded to the nearest thousand, how many units is that?

   A. 36,000

   B. 36,500

   C. 37,000

   D. 30,000

3. The souq sold 4,325 dates on Friday and 2,678 on Saturday. How many dates is that in all?

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4. A date farm has 6,000 dates and sells 2,457. How many dates are left?

   A. 4,457

   B. 3,643

   C. 3,553

   D. 3,543

5. Multiply: 23 x 14

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6. Four cousins share AED 852 equally. How many dirhams does each cousin get?

   A. 203

   B. 231

   C. 213

   D. 2,013

7. Which fraction is equal to 2/3?

   A. 4/6

   B. 2/6

   C. 3/4

   D. 4/9

8. Noura used 4/10 of a bag of flour on Monday and 5/10 of the bag on Tuesday. What fraction of the bag did she use?

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9. Omar drinks 2/5 of a litre of milk each day. How much milk is that in 3 days?

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10. Which decimal is equal to 3/10 + 4/100?

    A. 0.7

    B. 0.43

    C. 0.034

    D. 0.34

11. Layla has four ribbons that are 0.38 m, 0.4 m, 0.309 m and 0.299 m long. Which ribbon is the longest?

    A. 0.38 m

    B. 0.4 m

    C. 0.309 m

    D. 0.299 m

12. In the number 7,700, how many times as much is the first 7 worth as the second 7? Explain.

Answer guide for parents

Section 1: Patterns, equations and algebra thinking

1. 4 x 6 = 24
2. C. 45. 5 times as many means 5 x 9 = 45. (14 adds and 4 subtracts.)
3. AED 144 (bag AED 108, plus AED 36)
4. A. 4. 47 / 12 = 3 remainder 11, and those 11 students still need seats, so 4 buses.
5. 1 x 24, 2 x 12, 3 x 8, 4 x 6
6. B. 29. 29 has only the factors 1 and 29. The others have more: 27 = 3 x 9, 21 = 3 x 7 and 33 = 3 x 11.
7. Yes: 36 is a multiple of 9 because 9 x 4 = 36, so she fills 4 boxes.
8. 29, 35; add 6 each time.
9. D. 3, 6, 12, 24. 3 x 2 = 6, 6 x 2 = 12 and 12 x 2 = 24.
10. 26 (Sami 18, Ali 26)

Section 2: Shapes, measuring and space

1. C. 7 cm. Width = area / length: 56 / 8 = 7 cm.
2. 3,000 m
3. 150 minutes (2 x 60 + 30)
4. 55 cm (240 - 185)
5. 400 m (the perimeter is 2 x (120 + 80))
6. B. 90 degrees. From 12 to 3 is a quarter of a full turn, which is a right angle of 90 degrees.
7. D. 60 degrees. A straight line is 180 degrees, and 180 - 120 = 60 degrees.
8. 2
9. A. Right triangle. It has one angle of 90 degrees, so it is a right triangle.
10. Parallel

Section 3: Data, graphs and chance

1. B. 8. Each dot is one rope: 3 + 4 + 1 = 8.
2. 2/4 m (it has the most dots)
3. 2/4 m (3/4 - 1/4)
4. C. 2 m. 2 x 1/4 = 2/4 and 3 x 2/4 = 6/4, then 2/4 + 6/4 = 8/4 = 2 m.
5. 3 and 1/2 cm (3 x 1/2 = 1 and 1/2, and 2 x 1 = 2)
6. 4 (3 ribbons of 1/2 m and 1 ribbon of 3/4 m)
7. D. 1 and 1/8 litres. 3 x 1/8 = 3/8 and 2 x 3/8 = 6/8, then 3/8 + 6/8 = 9/8 = 1 and 1/8 litres.
8. 6
9. A. 1/3. 9/27 simplifies to 1/3 (divide the top and bottom by 9).
10. 3

Section 4: Numbers and calculating

1. Forty-five thousand, two hundred eight
2. A. 36,000. The hundreds digit is 4, which is less than 5, so round down to 36,000.

3. 7,003
4. D. 3,543. 6,000 - 2,457 = 3,543 (regroup across the zeros).
5. 322
6. C. 213. 852 / 4 = 213.
7. A. 4/6. Multiply the top and bottom by 2: 2/3 = 4/6.
8. 9/10
9. 6/5 = 1 and 1/5 litres (3 x 2/5)
10. D. 0.34. 3/10 = 30/100, and 30/100 + 4/100 = 34/100 = 0.34.
11. B. 0.4 m. 0.4 is 0.40, which is 40 hundredths. That is more than 38 hundredths.
12. 10 times: the first 7 is 7,000 and the second 7 is 700, and each place is worth 10 times the place to its right.